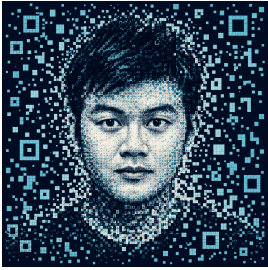


Bingru Huang

Curriculum Vitae



Assistant Researcher

School of Mathematical Sciences
University of Science and Technology of China (USTC)
Hefei, Anhui, China

Office: Room 515, Mathematics Building, East Campus

Email: hbr999@ustc.edu.cn

Homepage: <https://hbr1208.github.io/>

Google Scholar: Bingru Huang

Academic Appointments

Assistant Researcher

Current

School of Mathematical Sciences, University of Science and Technology of China

Postdoctoral Researcher

2023–2025

University of Science and Technology of China

Education

Ph.D. in Computational Mathematics

September 2017–June 2023

University of Science and Technology of China
School of Mathematical Sciences; Advisor: Prof. Falai Chen

B.Sc. in Computational Mathematics

September 2013–June 2017

Wuhan University of Technology

Research Interests

Mathematical foundations of splines and their applications in geometric design and numerical analysis:

- Isogeometric analysis;
- Approximation theory;
- Computer-aided geometric design;
- Mathematical theory of adaptive splines.

Teaching

Theory of Multivariate Function Approximation

Spring 2025

Graduate course; Instructor

Spline Functions and Approximation Theory

Fall 2024; Fall 2025

Graduate course; Instructor

Linear Algebra B1

Spring 2022

Teaching Assistant

Mathematical Analysis B1 & B2

2019–2021

Teaching Assistant

* Corresponding author.

8. **Bingru Huang**^{*} and Falai Chen, “A preliminary study on the dimensional stability classification of polynomial spline spaces over T-meshes,” *Computer Aided Geometric Design*, 130, 102609, 2026. DOI: 10.1016/j.cagd.2026.102609.
7. Shicong Zhong^{*}, **Bingru Huang**, and Falai Chen, “Modified HB-splines,” *Journal of Computational and Applied Mathematics*, 485, 117467, 2026. DOI: 10.1016/j.cam.2026.117467.
6. **Bingru Huang** and Falai Chen^{*}, “Dimension of bi-degree (d, d) spline spaces with the highest order of smoothness over hierarchical T-meshes,” *Science China Mathematics*, 69, 2195–2218, 2026. DOI: 10.1007/s11425-023-2499-9. arXiv: 2507.11047.
5. Shicong Zhong, **Bingru Huang**^{*}, and Falai Chen, “Basis construction for polynomial spline spaces over arbitrary T-meshes,” *Computer Aided Geometric Design*, 126, 102534, 2026. DOI: 10.1016/j.cagd.2026.102534. arXiv: 2508.12950.
4. Shicong Zhong, **Bingru Huang**, and Falai Chen^{*}, “Weighted Hierarchical B-Spline Basis,” *Journal of Computer-Aided Design & Computer Graphics*, 38(4), 731–741, 2026. [In Chinese] DOI: 10.3724/SP.J.1089.2024-00337.
3. **Bingru Huang** and Falai Chen^{*}, “An Algorithm to Compute the μ -Bases of Rational Parametric Surfaces with Respect to One Variable,” *Communications in Mathematics and Statistics*, 12(3), 523–541, 2024. DOI: 10.1007/s40304-022-00300-y.
2. **Bingru Huang** and Falai Chen^{*}, “On the stability of the dimensions of spline spaces with highest order of smoothness over T-meshes,” *Journal of Computational and Applied Mathematics*, 441, 115681, 2024. DOI: 10.1016/j.cam.2023.115681.
1. **Bingru Huang** and Falai Chen^{*}, “Computing μ -Bases of Univariate Polynomial Matrices Using Polynomial Matrix Factorization,” *Journal of Systems Science and Complexity*, 34(3), 1189–1206, 2021. DOI: 10.1007/s11424-020-9314-6.